



ChatGPT as an automated essay scoring tool in the writing classrooms: how it compares with human scoring

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Abstract

With the generative artificial intelligence (AI) tool's remarkable capabilities in understanding and generating meaningful content, intriguing questions have been raised about its potential as an automated essay scoring (AES) system. One such tool is ChatGPT, which is capable of scoring any written work based on predefined criteria. However, limited information is available about the reliability of this tool in scoring the different dimensions of writing quality. Thus, this study examines the relationship between the scores assigned by ChatGPT and a human rater and how consistent ChatGPT-assigned scores are when taken at multiple time points. This study employed a cross-sectional quantitative approach in analyzing 50 argumentative essays from each proficiency level (A2_0, B1_1, B1_2, and B2_0), totaling 200. These essays were rated by ChatGPT and an experienced human rater. Using correlational analysis, the results reveal that ChatGPT's scoring did not align closely with an experienced human rater (i.e., weak to moderate relationships) and failed to establish consistency after two rounds of scoring (i.e., low intraclass correlation coefficient values). These results were primarily attributed to ChatGPT's scoring algorithm, training data, model updates, and inherent randomness. Implications for writing assessment and future studies are discussed.

Keywords Generative AI · ChatGPT · Automated essay scoring · Automated writing evaluation · Argumentative essays

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1 Introduction

Recent developments in artificial intelligence (AI) and natural language processing (NLP) have given rise to state-of-the-art systems that are revolutionizing teaching, learning, and assessment (An et al., 2023; Ouyang et al., 2022; Weigle, 2013). Among these advancements is automated essay scoring (AES), which involves the process of assigning scores to student essays without human intervention, utilizing computer software or systems. This offers a potential solution to the time-consuming and resource-intensive task of grading essays and written assignments and ensures the reliable reporting of scores (Beseiso et al., 2021; Hussein et al., 2019; Zawacki-Richter et al., 2019). The potential of AES systems to transform the landscape of writing assessment has piqued the interest of scholars who aim to investigate their efficacy (e.g., Barrot, 2024a; Powers et al., 2015; Ramineni, 2013; Wilson et al., 2019). Some of these studies have compared AES to human raters to determine the reliability of the system (e.g., Dikli & Bleyle, 2014; Vo et al., 2023). This approach to establishing the reliability of AES has been considered as the gold standard, especially when compared with experienced raters (Powers et al., 2015).

Despite significant progress in AES in recent years, several issues have been identified. Most AES systems primarily rely on grammar, syntax, vocabulary, mechanics, and other form-related dimensions of writing for scoring (e.g., Grammarly and Project Essay Grader). Due to this heavy emphasis on linguistic dimensions, AES systems often fail to capture the relevance of ideas to the prompt, idea development, cohesion, and creativity, among other important aspects (Hussein et al., 2019; Ramesh & Sanampudi, 2022). While some extend their analysis to include ideas (e.g., e-rater and Intelligent Essay Assessor [IEA]), this analysis heavily relies on previously assessed essays and often lacks the depth needed to address content-related dimensions. For example, IEA evaluates ideas indirectly by assessing the substance of an essay in comparison to a collection of essays previously evaluated by proficient human raters, assigning a score based on how closely the content aligns with the essays in the training set.

With generative AI's remarkable capabilities in understanding and generating meaningful content, intriguing questions about its potential as an AES system have been raised. One such system is ChatGPT, which can score and provide qualitative feedback on any written work based on predefined criteria (e.g., Barrot, 2023). This innovative feature has drawn the attention of several L2 writing scholars in testing its efficacy for scoring essays (e.g., Mizumoto & Eguchi, 2023; Parker et al., 2023; Yancey et al., 2023). However, some inconsistencies in their findings raised questions on the reliability of ChatGPT in essay scoring and pointed out possible biases in the tool's scoring system. Moreover, little is known about whether ChatGPT can reliably and consistently provide scores for essays across various dimensions of writing quality, including content, substance, appropriateness for the audience, cohesion, and adherence to language style and conventions. Thus, this study investigates the reliability of ChatGPT as an AES tool and compares it with an experienced human rater. By delving into the capabilities

and limitations of ChatGPT as an AES system, we aim to shed light on the evolving landscape of AES and its implications for education and assessment practices, particularly in the field of second language (L2) writing. This study particularly seeks to determine the potential of ChatGPT in expediting the assessment process in writing classrooms and large-scale contexts while ensuring fair and consistent essay evaluation.

2 Literature review

2.1 Automated essay scoring

AES is a technology-driven approach that employs NLP techniques to score essays with perceived efficiency, objectivity, and consistency (Beseiso et al., 2021; Hussein et al., 2019; Zawacki-Richter et al., 2019). It resembles traditional human scoring, which assesses written texts using predetermined criteria. However, there are features that set AES apart from human rating (Barrot, 2024a; Ramineni & Williamson, 2013; Shermis, 2014). First, AES evaluates the various dimensions of writing instantaneously and comprehensively, allowing students more time for writing and teachers more time for instructional planning and delivery. Second, AES is less susceptible to scoring fatigue and the influence of personal biases, preferences, and preconceived notions, which can sometimes affect human scoring. Third, it is highly scalable and enables efficient processing and evaluation of a large volume of essays in a shorter timeframe, making it useful for large-scale assessments. Finally, this tool facilitates more transparent feedback for formative purposes. AES provides immediate and accessible feedback for students to use during the writing process, thereby enhancing learning and performance.

However, AES has some limitations. Among these is its lack of human nuances, as it struggles to capture the nuanced aspects of writing, such as creativity, style, reasoning, and emotional depth, which are critical in many contexts (Lee et al., 2023). AES also lacks contextual understanding, which may lead to the misinterpretation of the essay context, impacting the accuracy of its scoring (Ramesh & Sanampudi, 2022). Finally, AES is susceptible to gaming (Higgins & Heilman, 2014). For instance, students may attempt to “game” the system by using specific keywords or strategies to boost their scores rather than genuinely enhancing their writing skills.

Several comparative studies between AES and human scoring have been conducted in the field of writing, yielding valuable insights into the accuracy and reliability of AES. Some of these studies revealed a positive relationship between the two scoring approaches, while others reported otherwise. Among those who found a positive relationship between AES and human scoring is Shermis et al. (2002), who investigated the relationship between the scores assigned by Project Essay Grade (PEG) and six human raters. Their findings revealed that the interrater reliability among the six human raters, in combination with AES, was higher than the interjudge correlation among human raters alone, suggesting the efficiency of PEG for evaluating essays. In contrast, Wang and Brown (2008) found no significant relationship between AES (IntelliMetric) and human scoring within the context of

large-scale placement tests, particularly in terms of overall holistic scores and specific dimensions, namely focus, development, organization, and mechanics. However, a significant relationship was found in one dimension, sentence structure. Dikli and Bleyle (2014) also found discrepancies between AES (Criterion) and teacher feedback. They quantitatively and qualitatively analyzed AES and teacher feedback on 37 drafts and found that teachers provided a larger amount and better quality feedback on grammar, usage, and mechanics due to the fact that Criterion missed or misidentified several errors that students committed. More recently, Almusharraf and Alotaibi (2023) assessed the performance of AES (Grammarly) in comparison to human raters, examining 197 essays through error analysis and comparing the errors detected by AES and human raters. Although there was a moderate correlation between the two approaches, findings indicated that AES had a higher error detection rate and assigned lower scores to essays compared to human raters. The conflicting findings of these studies have raised additional questions regarding the validity and reliability of AES as a writing assessment tool.

2.2 ChatGPT as an L2 writing assessment tool

ChatGPT, which stands for Chat Generative Pre-trained Transformer, is an AI-driven conversational bot created by the American startup OpenAI. It leverages advanced computational methods and an extensive knowledge base to link words and concepts cohesively, allowing it to grasp prompts in their proper context (OpenAI, 2023). Powered by a Large Language Model (LLM), this tool possesses the capacity to comprehend and generate responses that resemble human language across a diverse array of subjects. One important element of LLM is randomness, which enables ChatGPT to produce somewhat divergent outcomes when the identical prompt is employed in various iterations or engagements (Schade, 2023). To ensure that ChatGPT is better equipped to respond to users' prompts, it undergoes frequent updates and fine-tuning that change its underlying algorithm and create different patterns and interpretations (Ray, 2023). Ray (2023) further explained that although not entirely devoid of biases, ChatGPT benefits from continuous endeavors to reduce biases in its training data.

ChatGPT provides a range of useful features as a language learning and teaching tool (Barrot, 2024b). With its vast reservoir of information and ability to mimic human writing, it serves as a valuable and highly accessible resource for language input and practice. It can also remember previous conversations, enabling users to ask follow-up questions and sustain coherent and fluent conversations. As an automated writing assistant, this AI tool can generate topics across disciplines, identify the writing context and audience, engage in brainstorming, create an outline, paraphrase a text, and produce coherent and linguistically correct texts, which can serve as model texts for writing (Dergaa et al., 2023; Su et al., 2023).

Another key affordance of ChatGPT is its ability to provide automated written corrective feedback (AWCF), which is feedback provided by automated writing evaluation (AWE) tools about users' written output (Barrot, 2023; Han & Sari, 2024). This feature addresses the need of writers for timely and quality feedback. ChatGPT

provides qualitative feedback on a given text when prompted. To illustrate, ChatGPT lists detailed comments on content, clarity, structure, idea development, specificity, and stylistic conventions when a user types the prompt “offer feedback on the following text.” More specifically, it provides both the strengths and weaknesses of the papers and recommends possible revisions that the user can adopt to further improve the paper. The generated criteria and comments are based on the corpus processed by ChatGPT as a massive collection of patterns and associations. The feedback that ChatGPT generates becomes more specific and detailed when predefined criteria with corresponding descriptions are provided. These may include, but are not limited to, depth of discussion, relevance to the prompt, clarity of purpose and audience, cohesion, idea development, structure and organization, grammar and mechanics, and language style. This tool is also capable of assigning holistic and analytic scores to written texts based on certain criteria and score ranges. Along with these scores are explanations that justify them. These AWCF features are useful for both students and teachers for formative and summative assessment purposes. However, it is not clear whether these scores are valid and reliable and can mimic experienced human raters.

Studies on the utilization of ChatGPT for L2 writing have recently been building up (e.g., Athanassopoulos et al., 2023; Javier & Moorhouse, 2023; Marzuki et al., 2023; Su et al., 2023). However, little research is available about how ChatGPT can be tapped as an L2 writing assessment tool. One of them is Guo and Wang’s (2024) research, which examined ChatGPT’s performance in generating feedback on English as a foreign language (EFL) students’ argumentative writing. Findings revealed that ChatGPT produced a statistically larger amount of feedback compared to the feedback given by five EFL teachers. This AI tool also distributed its attention relatively equally among content, organization, and language, unlike teachers who mainly focused on content- and language-related issues. As for the teachers’ perception of ChatGPT as a feedback generator, they felt the tool was capable of assessing essays, could provide encouraging feedback, and was specific and flexible. They added that this AI tool could serve as a supplement and lessen teachers’ burden of assessing student essays. However, they also reported some challenges, such as lengthy, unreadable, and irrelevant comments, lack of knowledge of the teaching context, and accessibility. A similar study was undertaken by Schmidt-Fajlik (2023), who compared ChatGPT with Grammarly and ProWritingAid as automated grammar checkers. His findings indicated that first-year Japanese students positively accepted the three tools, and they have the potential to improve their writing accuracy. He added that while Grammarly and ProWritingAid offer metalinguistic explanations, ChatGPT has a more intuitive and straightforward visual interface and offers more detailed and precise feedback. Schmidt-Fajlik’s (2023) work shed light on ChatGPT’s potential as a language editor and AWCF tool but was limited only to the linguistic aspect of a written text.

Mizumoto and Eguchi’s (2023) further advanced this area by going beyond the language editing affordance of ChatGPT. Their research assessed 12,100 essays using the GPT-3.5 model and found that the tool demonstrated a certain degree of accuracy and reliability in essay scoring. However, the scoring was confined to linguistic features. This data suggested that the inclusion of linguistic features could improve the accuracy of essay scoring with ChatGPT. Yancey et al. (2023) expanded the analysis

by comparing the effectiveness of GPT-3.5 and GPT-4 in evaluating essays based on both linguistic and non-linguistic criteria. By using human raters as a benchmark, they found that ChatGPT-4 nearly matched the performance of modern automated writing evaluation methods. However, when compared to human ratings, the agreement varied depending on the test-taker's first language. Similar results were reported by Parker et al. (2023), who examined 42 essays produced by nursing students. They reported that ChatGPT demonstrated viability as an AES and stricter scoring than human raters.

One plausible explanation for the variations in findings across the aforementioned studies lies in the diverse nature of automated feedback examined. Guo and Wang (2024) and Schmidt-Fajlik (2023) focused on ChatGPT's qualitative feedback, emphasizing its robustness as an AWE system. In contrast, Yancey et al. (2023), Parker et al. (2023), and Mizumoto and Eguchi (2023) concentrated on ChatGPT's scoring capability, revealing its limitations in providing quantitative feedback. Due to limited research exploring the scoring aspect, it remains uncertain whether ChatGPT is valuable for quantitative writing assessment. Thus, additional insights are necessary to gauge its potential as an AES system. Although Yancey et al. (2023), Parker et al. (2023), and Mizumoto and Eguchi (2023) delved into this domain, the examination did not cover the assessment of score consistency across different time points, leaving uncertainties about the reliability of ChatGPT's scoring ability. This gap raises critical questions regarding ChatGPT's reliability in generating scores, particularly at the per-criterion level. While preliminary information suggests ChatGPT can assign scores to essays (Barrot, 2023), the validity and accuracy of these scores remain unclear. Given the potential impact on L2 writing pedagogy and assessment practices, a thorough empirical investigation is warranted.

2.3 Research questions

To address the gaps in the literature, this study investigates the reliability of ChatGPT as an AES tool and compares its scoring with human raters. It aims to advance our understanding of the capabilities of a generative AI tool in assessing the multidimensionality of L2 writing performance with precision. Hence, we have formulated the following research questions:

1. What is the relationship between the scores assigned by ChatGPT and human raters?
2. How consistent are the scores assigned by ChatGPT at multiple time points compared to those assigned by human raters?

3 Materials and methods

3.1 Data source

This study employed a cross-sectional quantitative approach to address the research questions. All argumentative essays were obtained from the ICNALE-Written

corpus, which is available at <http://language.sakura.ne.jp/icnale/download.html>. This corpus contains 5,600 essays written by college students from ten Asian countries/territories, namely China, Hong Kong, Indonesia, Japan, Pakistan, the Philippines, Singapore, South Korea, Taiwan, and Thailand. These essays were categorized based on the students' proficiency levels, which include A2_0 (waystage), B1_1 (threshold, lower), B1_2 (threshold, upper), and B2_0 (vantage or higher) (Ishikawa, 2013). Fifty (50) essays were randomly selected from each proficiency level (i.e., 5 from each country), resulting in a total of 200 essays. Analyzing essays from various proficiency levels allows researchers to assess the generalizability of ChatGPT's effectiveness across different language learning stages. In the same way, these essays were drawn from ten different countries to provide a balanced and comprehensive overview of the efficacy of ChatGPT across linguistic and cultural backgrounds. Each essay used a similar prompt (i.e., *it is important for college students to have a part-time job*), comprised 200 to 300 words, and was written using a word processor.

3.2 Procedure

We adopted the Common Core State Standards (CCSS)-aligned writing rubric, which was used by the human rater and ChatGPT (GPT-3.5 version) in assessing the student essays. This rubric has three versions to evaluate different text types, namely narrative, argumentative, and informative. For this study, we adopted the rubric for argumentative essays, which covers five criteria: claim, development, audience, cohesion, and style and conventions. *Claim* signifies that the text introduces a clear, debatable assertion substantiated by valid reasons and supporting evidence. Regarding *development*, it entails the text furnishing ample data and evidence to substantiate the claim, while also conscientiously acknowledging the strengths and limitations inherent in both the claim and counterclaim. In relation to *audience*, it pertains to the text's ability to anticipate the audience's knowledge level, concerns, values, and potential biases concerning the claim, thereby addressing the audience's specific needs effectively. Concerning *cohesion*, it denotes the text's adept utilization of words, phrases, clauses, and varied sentence structures to establish seamless links within the text, fostering cohesion and elucidating the connections between the claim and reasons, reasons and evidence, as well as claims and counterclaims. Lastly, *style and conventions* necessitates that the text upholds a formal, objective tone, showcasing adherence to the standard conventions of English usage, mechanics, and documentation style. Each criterion is assigned the following scores: 1 (inadequate), 2 (developing), 3 (proficient), 4 (skilled), and 5 (exceptional). Each score level has a corresponding description related to a specific criterion.

In scoring each essay with ChatGPT, we instructed the AI tool to assess each essay using the CCSS-aligned writing rubric and provide a score for each criterion. Our instructions consisted of typing, "Rate the essays using these criteria. Give a score per criterion." This was followed by a detailed presentation of the criteria. Below is an excerpt of these instructions.

Rate the essays using the following criteria. Give a score per criterion. The writing prompt is: It is important for college students to have a part-time job

Criterion 1: Claim. The text introduces a clear, arguable claim that can be supported by reasons and evidence

- 5: The text introduces a compelling claim that is clearly arguable and takes a purposeful position on an issue. The text has a structure and organization that is carefully crafted to support the claim
 - 4: The text introduces a precise claim that is clearly arguable and takes an identifiable position on an issue. The text has an effective structure and organization that is aligned with the claim
 - 3: The text introduces a claim that is arguable and takes a position. The text has a structure and organization that is aligned with the claim
 - 2: The text contains an unclear or emerging claim that suggests a vague position. The text attempts a structure and organization to support the position
 - 1: The text contains an unidentifiable claim or vague position. The text has limited structure and organization
-

One by one, we copied each essay from the MS Word file, pasted it into the text-box, and clicked the “proceed” button. It took ChatGPT 8 to 12 s to evaluate each essay. Following the evaluation, the AI tool provided scores for each criterion, along with corresponding explanations, and offered an overall impression of the essay. To assess ChatGPT’s consistency in providing scores, we subjected each essay to a second round of ChatGPT scoring using the same prompt four weeks after the initial round. The first round of scoring occurred from August 29, 2023 to September 2, 2023, while the second round took place from September 26 to 30, 2023. Below is an example of the scores provided by ChatGPT.

Let’s evaluate the essay using the provided criteria:

Criterion 1: Claim

Score: 3

The essay asserts that part-time jobs have both positive and negative sides for college students. It suggests that part-time jobs can add experience but may also lead to laziness and negative traits. The claim is clear and effectively presents both sides of the argument

Criterion 2: Development

Score: 2

The essay briefly develops the idea that part-time jobs can add experience and may foster perseverance. However, the development of the negative aspects, such as laziness and negative traits, is somewhat limited and lacks specific examples or evidence to support the points

Criterion 3: Audience

Score: 3

The essay appears to address an audience interested in the topic of whether college students should have part-time jobs. It presents both positive and negative aspects, which could engage a reader with varying perspectives

Criterion 4: Cohesion

Score: 2

The essay has a relatively clear structure and organization but could benefit from smoother transitions between ideas. The connection between the positive and negative aspects of part-time jobs could be improved for better coherence

Criterion 5: Style and Conventions

Let's evaluate the essay using the provided criteria:

Score: 3

The essay uses generally appropriate language and demonstrates a reasonable level of grammar and usage. While there are a few awkward phrasings, they do not significantly affect the overall clarity of the essay

Overall, this essay presents both positive and negative aspects of part-time jobs for college students but could benefit from more thorough development and specific examples to support the points. The organization and transitions could be improved for better coherence

A single trained and experienced human rater scored all 200 essays using the aforementioned writing rubric. This rater holds a PhD and possesses 20 years of experience in teaching English and assessing English essays, both at the undergraduate and graduate levels. Additionally, he has served as an essay rater for a well-known educational testing organization and an international learner corpus of written and spoken texts. Using a single experienced rater ensures consistency in scoring. Variability in human scoring can occur when multiple raters are involved due to differences in their interpretations and application of the scoring criteria. By employing one experienced rater, the study minimizes this variability and maintains a uniform standard for comparison with ChatGPT.

Before scoring each essay, he thoroughly familiarized himself with the criteria. He conducted a pilot scoring of 25 essays, repeating this process twice with a three-week gap between the two sets of essay scoring to ensure that the rater consistently applied the scoring criteria. Note that the human rater did not have access to the previous score he gave to the same writing when scoring for the second time. The results of the correlation coefficients indicate that the human rater consistently scored the essays across all five criteria ($r=0.7538$ for claim, $r=0.7592$ for development, $r=0.8395$ for audience, $r=0.8879$ for cohesion, $r=0.8909$ for style and conventions, and $r=0.9108$ for the overall score). After this, he proceeded to the actual rating of the 200 essays, with each evaluation taking approximately 15 min. Note that the human rater did not have access to information about proficiency level to avoid any unnecessary influence in his scoring. Subsequently, all scores from ChatGPT and the human rater were tabulated in a spreadsheet for data analysis. This procedure was done in accordance with ethical research practices, including but not limited to obtaining informed consent from the human rater, ensuring confidentiality and anonymity, and avoiding harm.

3.3 Data analysis

After tabulating all the scores from both ChatGPT and the human rater, we subjected them to both descriptive and inferential statistics. To calculate the overall score for each essay, we simply added the scores from the five criteria, both for ChatGPT and the human rater. We employed Pearson's correlation coefficient to measure the statistical relationship between the scores assigned by ChatGPT and the human rater. The interpretation of the correlation was as follows: 0.00–0.10 as negligible, 0.10–0.39 as weak, 0.40–0.69 as moderate, 0.70–0.89 as strong, and 0.90–1.00 as

very strong (Schober et al., 2018). To address research question 2, we utilized the intraclass correlation coefficient (ICC), specifically the single-measure, single-rater reliability, to assess the consistency of ChatGPT-assigned scores at different points in time. For this analysis, we considered both the overall scores and scores per criterion. The following ICC interpretation was applied (Liljequist et al., 2019): <0.50 – poor; 0.50 to 0.75 – moderate; 0.75 to 0.90 – good; above 0.90 – excellent.

4 Results

The present study investigated the efficacy of ChatGPT as an AES system. Specifically, we aimed to establish the correlation between the scores assigned by ChatGPT and human raters and to assess ChatGPT's consistency in assigning scores across multiple time points. To achieve these objectives, we employed a combination of descriptive and reliability statistics.

Table 1 summarizes the results of the two rounds of scoring conducted by ChatGPT and a human rater. The data indicates that the mean scores assigned by the human rater are relatively higher across all criteria and for the overall score when compared to the scores assigned by ChatGPT. This pattern is consistently observed in both rounds, with greater score variability noted in the second round of scoring. Among the five criteria, ChatGPT assigns the highest and lowest mean scores to the *claim* and *cohesion*, respectively, in both rounds. Conversely, the human rater, on average, assigns the highest score to *style* and the lowest score to *development*. Upon closer examination of the mean scores by proficiency level, the data reveals that the scores assigned by the human rater consistently increase linearly with proficiency level in both rounds of scoring (A2_0=12.90, B1_1=14.96, B1_2=16.96, and B2_0=22.15 for round 1; A2_0=13.10, B1_1=15.08, B1_2=17.13, and B2_0=22 for round 2). In contrast, ChatGPT's mean scores across proficiency levels appear inconsistent for round 1 (A2_0=13.94, B1_1=14.13, B1_2=14.04, and B2_0=16.69) and round 2 scoring (A2_0=11.54, B1_1=12.67, B1_2=15.15, and B2_0=14.40). These results suggest that, unlike ChatGPT, the human rater was able to capture the nuances in the essays produced by writers with different proficiency levels.

Table 2 presents the relationship between the scores assigned by ChatGPT and the human rater. The results reveal a weak relationship between the two sets of scores across the five criteria (ranging from $r=0.259$ to $r=0.358$ for round 1, and from $r=0.172$ to $r=0.393$ for round 2) as well as in the overall score ($r=0.359$ for round 1 and $r=0.382$ for round 2). This level of relationship consistency was observed in both rounds of scoring. The weak relationship suggests that ChatGPT's scoring does not closely align with the assessments made by the human rater. This lack of consistency raises concerns about the reliability of ChatGPT and its accuracy in capturing the same aspects of writing quality or criteria as the human rater.

We used ICC to determine the consistency between ChatGPT-assigned scores taken at a three-week gap. As presented in Table 3, ChatGPT scores generated at two time points show poor reliability in the overall score ($r=0.447$) and across the

Table 1 Scores assigned by ChatGPT and Human Rater

	ChatGPT				Human Rater			
	Round 1		Round 2		Round 1		Round 2	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Claim	3.23	0.77						
Development	2.87	0.90	2.89	0.74	3.36	0.86	3.33	0.84
Audience	3.01	0.82	2.56	0.77	3.26	0.96	3.28	0.89
Cohesion	2.78	0.87	2.79	0.78	3.35	0.93	3.35	0.93
Style	2.81	0.74	2.55	0.74	3.31	1.02	3.33	0.99
Overall Score	14.69	3.76	2.61	0.77	3.35	1.16	3.42	1.12
			13.39	3.46	16.63	4.60	16.70	4.38

Table 2 Correlations between the scores assigned by ChatGPT and the Human Rater

	Round 1	Round 2
Claim	0.259*	0.375*
Development	0.328*	0.172*
Audience	0.263*	0.362*
Cohesion	0.311*	0.323*
Style	0.358*	0.393*
Overall Score	0.359*	0.382*

*Correlation is significant at the 0.05 level (2-tailed)

Table 3 Intraclass correlations of ChatGPT-assigned scores and Human-assigned scores

	ChatGPT	Human Rater
Claim	0.386*	0.917*
Development	0.394*	0.894*
Audience	0.295*	0.922*
Cohesion	0.385*	0.938*
Style	0.481*	0.965*
Overall Score	0.447*	0.973*

*Correlation is significant at the 0.05 level (2-tailed)

five criteria ($r=0.295$ to $r=0.481$). Low ICC values indicate that ChatGPT scoring was not producing stable or reliable scores. This inconsistency means that the tool's scoring varied widely, making it challenging to trust its results. Unlike ChatGPT, the scores assigned by the human rater received good to excellent reliability, suggesting the stability and trustworthiness of the results. All criteria show excellent reliability ($r=0.917$ for claim, $r=0.922$ for audience, $r=0.938$ for cohesion, and $r=0.965$ for style), except for development which obtained good reliability ($r=0.894$). In the same way, the overall scores given by the human rater at two different instances received excellent reliability ($r=0.973$).

5 Discussion

This study aimed to examine ChatGPT's efficacy as an AES tool and compare it to human scoring. The overall results suggest that ChatGPT has a limited capacity for automated scoring. As the data shows, this AI tool's scoring did not closely align with an experienced human rater and failed to establish consistent scoring. Our study extends earlier reports on discrepancies between human scoring and feedback with AES (Almusharraf & Alotaibi, 2023; Dikli & Bleyle, 2014; Wang & Brown, 2008) and ChatGPT (Guo & Wang, 2024; Schmidt-Fajlik, 2023; Yancey et al., 2023). It also confirms Almusharraf and Alotaibi's (2023) report on AES's tendency to assign lower scores than human raters. Unlike these previous studies that exclusively focused on comparing ChatGPT with human raters, we attempted to unravel

the ability of this tool to generate consistent scoring across different time points. Data indicates that ChatGPT fell short in this aspect, raising doubts about its trustworthiness and dependability.

Several factors may have contributed to the weak relationship between the scores assigned by ChatGPT and the human rater. First, ChatGPT's scoring algorithm may not be advanced enough to fully grasp and comprehensively assess complex aspects of writing. Because of this, ChatGPT and the human rater might have interpreted and applied the scoring criteria differently. While human raters can distinguish unconventional writing approaches, creativity, and subtlety in the argumentation of human-produced essays, ChatGPT might have struggled with these aspects of writing (Yancey et al., 2023). Hence, ChatGPT may potentially miss key points or misinterpret the writer's intent. The training data used to teach ChatGPT is another pressing concern. Training data refers to the datasets used to fine-tune or train a machine learning model to make predictions. It is likely that ChatGPT's training data does not yet fully represent and cannot fully capture the range of writing quality, such as style, substance and depth, organization and idea development, cohesion, and intent. It may also inadvertently contain biases that led to differences in scores. As Ray (2023) explained, generative AI tools, such as ChatGPT, potentially inherit biases from their training data. Additionally, our study revealed that ChatGPT assigned relatively lower scores compared to the human rater. Although the AI tool may not be fully trained yet to assess writing nuances, it has the ability to detect errors more comprehensively and can focus on many aspects of writing simultaneously (Guo & Wang, 2024; Schmidt-Fajlik, 2023). In this regard, we resonate with Parker et al. (2023) and Almusharraf and Alotaibi (2023), who conjectured that AES's capability to identify a wide range of errors led to its assigning lower scores than human raters. In the case of the experienced human rater, the data indicates that he was more permissive about the errors considering his experience in scoring essays produced by non-native speakers of English.

An important finding of this study is the inconsistent scores provided by ChatGPT at two different points in time. Several possible explanations can account for this inconsistency. Given the three updates (September 11, 25, and 27, 2023) made in ChatGPT between the two rounds of scoring, we might construe that these model updates and fine-tuning led to ChatGPT exhibiting variations in its scoring because of the changes in its underlying algorithm and processed corpus. It is important to understand that when new data is introduced to ChatGPT, it may create different patterns and interpretations (Ray, 2023). This phenomenon is known as model drift (Gonzalez Torres & Sawhney, 2023; Suppadungsuk et al., 2023). Despite these updates, ChatGPT still failed to match the rating ability of an experienced human rater. Suppose the updates did not significantly alter the way ChatGPT assigns scores. In that case, another factor that might have contributed to the intra-class inconsistency is the inherent randomness of ChatGPT responses. Randomness is an element of LLM that allows it to yield slightly different results when the same prompt is used in different runs or interactions (Schade, 2023).

Our study identified potential weaknesses in ChatGPT's algorithm that may have influenced its reliability. The tool struggled to fully grasp and evaluate complex writing aspects such as creativity and subtle argumentation, leading to limitations in

interpreting and applying scoring criteria comprehensively. The training data did not fully capture the range of writing quality and led to potential biases and inaccuracies. Generative AI tools like ChatGPT can inherit biases from their training data, which influence scoring outcomes and potentially lead to unreliable assessment. ChatGPT also tended to assign lower scores than human raters due to its comprehensive error detection capabilities, which may undervalue the overall quality and intent of writing. Qualitatively, a human rater brought a nuanced understanding that includes appreciating creativity and unconventional approaches, while ChatGPT focused heavily on technical correctness. A human rater could balance technical accuracy with rhetorical effectiveness, whereas ChatGPT might have penalized technical errors more heavily. Model updates between scoring rounds led to variations in ChatGPT's scoring patterns. This randomness in ChatGPT's responses could have resulted in different scoring outcomes for the same essay at different times, challenging consistent and reliable scoring practices.

The study's results are generalizable due to the diverse data sources used, including essays from college students across ten Asian countries and various proficiency levels. This diversity ensures that the results are representative of different linguistic and cultural backgrounds. Analyzing essays from different proficiency levels provided insights into ChatGPT's scoring performance across various stages of language learning. While the study focused on L2 writing in college settings, the findings can be extrapolated to other educational levels and contexts, although further research is needed to validate the tool's effectiveness in different environments and with diverse student populations. These insights provided a foundation for understanding the potential and limitations of ChatGPT as an AES tool and inform future developments and applications in writing instruction and assessment.

6 Conclusion

This study examined the efficacy of ChatGPT as an AES system by comparing it to an experienced human rater. Overall findings indicate that this AI tool's capabilities in automated scoring are highly constrained. As shown, ChatGPT's scoring did not closely match that of an experienced human rater and lacked the ability to maintain consistent scoring due to various factors, such as ChatGPT's scoring algorithm, training data, model updates, and inherent randomness. These findings have a far-reaching impact on the trustworthiness, fairness, and effectiveness of assessment performed by the tool. Our study also underscores the potential and challenges of integrating ChatGPT and similar tools in writing assessment, while highlighting the need for further research in this rapidly developing field. Nonetheless, it is important to note that ChatGPT is continuously being updated, and evidence indicates that its future versions may be better equipped with automated scoring.

Several implications can be drawn from the findings of our study. First, this study highlights the limitations of the current version of ChatGPT and opens new avenues for exploring how AI models can address these limitations. One way is to recalibrate the training data and scoring algorithm, allowing ChatGPT to distinguish quality from poorly written texts. Second, while ChatGPT can expedite the assessment

process, it does not fully capture the multidimensionality of writing. Human raters may still be necessary for final evaluations, especially for essays from non-native speakers, to capture the nuances and creativity in student writing. Hence, this tool should be used with careful consideration in writing classrooms and large-scale contexts. Finally, the utilization of ICC enhances methodological robustness by going beyond the relationships between the scores assigned by human raters and ChatGPT (or other similar AES systems). In this study, we also looked into the intraclass consistency, which sets a precedent for future research examining this line of research. Our study also has implications for writing instruction and pedagogical practices, beyond the assessment context. First, the feedback mechanism of ChatGPT can help scaffold the writing process and promote self-regulation among students. Second, training teachers to effectively incorporate ChatGPT into their instruction can lead to more innovative teaching practices. Understanding the limitations and strengths of ChatGPT can help them better guide students in using these technologies responsibly. Finally, teachers can use ChatGPT's feedback to tailor their instruction to meet the individual needs of students, providing more targeted support for those at different stages of language learning.

While this study has contributed valuable insights, it is essential to recognize its limitations. First, our study primarily compared ChatGPT scores with those of a single experienced human rater. A more comprehensive analysis could involve several experienced human raters to limit or eliminate human-rater factors, thus providing a more nuanced understanding of ChatGPT's efficacy. Second, this study exclusively examined the scores assigned by ChatGPT and did not consider its qualitative feedback. While the scores are a critical aspect of assessing essays, excluding the qualitative comments restricts the comprehensive evaluation of the essays and their interpretation. Examining the validity and reliability of qualitative feedback solely or in combination with scores warrants further investigation for a more holistic evaluation of ChatGPT's potential as an AWE tool. Finally, future studies may explore the potential influence of student-level factors, such as language proficiency and writing instruction, on the performance of both ChatGPT and human raters. This information could provide a more holistic understanding of the assessment process.

There are compelling reasons to believe that ChatGPT will improve in the future. Continuous learning and the influx of diverse data will enrich its training, while advancements in NLP algorithms will enhance its understanding of complex writing aspects. Enhanced training techniques, such as transfer and reinforcement learning, will enable more effective learning from smaller datasets. Efforts to address biases will lead to fairer and more objective scoring, and integrating human feedback will align ChatGPT's scoring with human judgment. Future versions of ChatGPT will be more customizable and adaptable to specific contexts, ensuring more accurate evaluations. Robust error detection and correction mechanisms will provide nuanced feedback, and broader acceptance and integration in education will drive further investment in research and development. Collectively, these factors will significantly improve ChatGPT's accuracy, reliability, and effectiveness in the future.

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Data availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval The study has undergone appropriate ethics protocol.

Consent to participate Informed consent was sought from the participants.

Consent for publication Authors consented the publication. Participants consented to publication as long as confidentiality is observed.

Competing interests None

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